Yinda Zhang

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Education

University of Pennsylvania — **Ph.D., Computer and Information Science**
Advisor: Prof. Vincent Liu — *Sep. 2022 - Present*

University of Chicago — **Pre-Doctoral M.S., Computer Science**
Advisor: Prof. Junchen Jiang — *Sep. 2020 - Dec. 2021*

Peking University — **B.S., Computer Science and Technology**
Advisor: Prof. Tong Yang — *Sep. 2016 – Jul. 2020*

Publication

Conference

1. **Yinda Zhang**, Liangcheng Yu, Gianni Antichi, Ran Ben Basat, Vincent Liu, Enabling Silent Telemetry Data Transmission with InvisiFlow, USENIX Symposium on Networked Systems Design and Implementation (*NSDI 2025*)

2. **Yinda Zhang**, Peiqing Chen, Zaoxing Liu, OctoSketch: Enabling Real-Time, Continuous Network Monitoring over Multiple Cores, USENIX Symposium on Networked Systems Design and Implementation (*NSDI 2024*)

3. Yikai Zhao*, Wenchen Han*, Zheng Zhong*, **Yinda Zhang**, Tong Yang, Bin Cui, Double-Anonymous Sketch: Achieving Fairness for Finding Global Top-K Frequent Items, ACM International Conference on Management of Data (*SIGMOD 2023*)

4. Yikai Zhao*, **Yinda Zhang***, Yuanpeng Li*, Yi Zhou, Chunhui Chen, Tong Yang, Bin Cui, MinMax Sampling: A Near-optimal Global Summary for Aggregation in the Wide Area, ACM International Conference on Management of Data (*SIGMOD 2022*)

5. Zhuochen Fan, **Yinda Zhang**, Tong Yang, Mingyi Yan, Gang Wen, Yuhan Wu, Hongze Li, Bin Cui, PeriodicSketch: Finding Periodic Items in Data Streams, IEEE International Conference on Data Engineering (*ICDE 2022*)

6. Tong Yang, Jizhou Li, Yikai Zhao, Kaicheng Yang, Hao Wang, Jie Jiang, **Yinda Zhang**, Nicholas Zhang, QCluster: Clustering Packets for Flow Scheduling, International World Wide Web Conference (*WWW 2022*)

7. **Yinda Zhang**, Zaoxing Liu, Ruixin Wang, Tong Yang, Jizhou Li, Ruijie Miao, Peng Liu, Ruwen Zhang, Junchen Jiang, CocoSketch: High-Performance Sketch-based Measurement over Arbitrary Partial Key Query, Annual conference of the ACM Special Interest Group on Data Communication (*SIGCOMM 2021*)

8. **Yinda Zhang**, Jinyang Li, Yutian Lei, Tong Yang, Zhetao Li, Gong Zhang, Bin Cui, On-Off Sketch: A Fast and Accurate Sketch on Persistence, International Conference on Very Large Data Bases (*VLDB 2021*)

9. Xiangyang Gou*, Long He*, **Yinda Zhang***, Ke Wang, Xilai Liu, Tong Yang, Yi Wang, Bin Cui, Sliding Sketches: A Framework using Time Zones for Data Stream Processing in Sliding Windows, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (*KDD 2020*)

Journal

1. Ruijie Miao*, **Yinda Zhang***, Zihao Zheng*, Ruixin Wang, Ruwen Zhang, Tong Yang, Zaoxing Liu, Junchen Jiang, High-Performance Sketch-based Measurement over Arbitrary Partial Key Query, IEEE/ACM Transactions on Networking (*TON 2023*)

2. Xiangyang Gou*, **Yinda Zhang***, Zhoujing Hu*, Long He, Ke Wang, Xilai Liu, Tong Yang, Yi Wang, Bin Cui, A Sketch Framework for Approximate Data Stream Processing in Sliding Windows, IEEE Transactions on Knowledge and Data Engineering (*TKDE 2022*)

SELECTED RESEARCH PROJECTS

Programmable Networks

Collecting Network Telemetry Without Affecting User Traffic — *Nov. 2022 – Sep. 2024*

- Developed a communication substrate for silently collecting network telemetry from switches
- Implemented the system on the *Tofino switch* and conducted large-scale simulations on *ns-3*
- Achieved near-zero overhead on user traffic while maintaining high sustainable throughput for telemetry data

Real-Time, Continuous Network Monitoring over Multiple Cores — *Sep. 2021 – Sep. 2022*

- Designed a scalable software framework for sketch-based monitoring across multiple cores
- Applied the framework to nine sketch algorithms on three platforms: *CPU, DPDK, and XDP*
- Improved accuracy by 15.6× and throughput by 4.5× over state-of-the-art solutions

High-Performance Measurement over Arbitrary Partial Key Query — *Jul. 2020 – Jul. 2021*

- Developed a measurement system supporting simultaneous queries on multiple keys
- Implemented the system on *CPU, Open vSwitch, P4, and FPGA* platforms
- Enhanced accuracy by 10.4× and throughput by 27.2× in multi-key flow measurement

Approximate Streaming Algorithm

Near-optimal Global Summary for Aggregation in Wide Area Networks — *Jun. 2021 – Mar. 2022*

- Designed a fast, adaptive, and accurate sampling scheme for global aggregation in WAN
- Derived error bounds and proved the optimality of the algorithm
- Applied the algorithm to *federated learning, distributed state aggregation, and hierarchical aggregation*

Fast and Accurate Sketch for Persistence Estimation — *Jan. 2020 – Sep. 2020*

- Designed a sketch algorithm for persistence estimation and identifying persistent items
- Derived error bounds and demonstrated reduced space complexity
- Achieved 6.17× better accuracy across multiple datasets in experimental evaluations

INDUSTRY EXPERIENCE

ByteDance | Applied Research Intern — Bellevue, Washington

Hardware Acceleration Architectures for Layer-4 Load Balancing — *May 2024 – Present*

- Compared two hardware acceleration architectures for stateful layer-4 load balancing.
- Conducted in-depth analysis of the design of *AMD Pensando DPU accelerators*

High-Performance Flow Logging via Matrix Sketching — *May 2024 – Present*

- Designed a flow logging system generates approximate summaries and supports diverse query granularities
- Leveraged *matrix sketching* theory and implemented the system on the *Tofino switch*

Conviva | **Research Intern** **Beijing, China**

Online Profiling and Adaptation of Quality Sensitivity for Internet Video *Jul. 2020 – Jan. 2021*

- Designed and implemented a data processing engine for time-state analytics
- Measured viewership patterns across 7.6M video sessions from four content providers
- Conducted in-depth analysis of the relationship between video content and quality sensitivity

MISC

Languages: C++/C, Python, CUDA, Java, SQL
Fellowship: Jonathan M. Smith Fellowship (2022)
Teaching Assistant: CIS 553: Networked Systems (Spring 2023, Fall 2023)
Shadow TPC Member: IMC 2022, EuroSys 2023